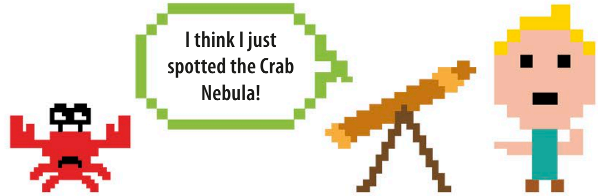
Starry sky

The next steps will wrap up your star as a Python function. You'll then be able to use that function to draw a sky that's teeming with stars.



The **draw_star()** function uses five parameters to define the shape, size, colour, and position of the star.

9 Create the star function

Edit the code as shown here. It replaces nearly all of your existing code with a new version. The large block wraps up all the star-drawing instructions and keeps them neatly together as a function. You can now use this function to draw a star in your main code with a single line of Python, **draw_star()**.

```
import turtle as t

def draw_star(points, size, col, x, y):
    t.penup()
    t.goto(x, y)
    t.pendown()
    angle = 180 - (180 / points)
    t.color(col)
    t.begin_fill()
    for i in range(points):
        t.forward(size)
        t.right(angle)
    t.end_fill()

# Main code
t.Screen().bgcolor('dark blue')
draw_star(5, 50, 'yellow', 0, 0)
```

The x and y coordinates set the position of the star on the screen.

This sets the background colour to dark blue.

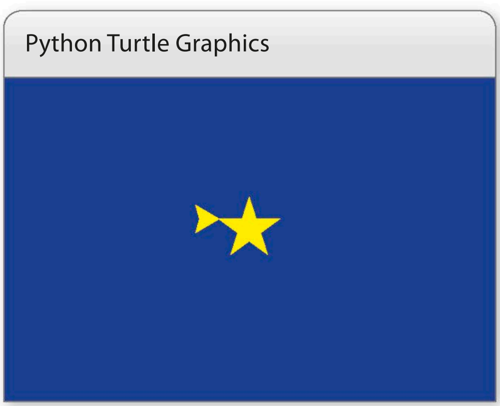
This "comment" line starting with a hash symbol (#) isn't part of the code run by Python. It's like a label to help you understand the program.

This line calls (runs) the function.

The turtle draws a yellow, five-pointed star, size 50, in the centre the window.

10 Run the project

The turtle should draw a single yellow star on a blue background.



REMEMBER

Comments

Programmers often put comments in their code to remind them what different parts of a program do or to explain a tricky part of a project. A comment must start with a #. Python ignores anything you type on the same line after the # and doesn't treat it as part of the code. Writing comments in your own projects (such as the line **# Main code** shown above) can be really helpful when you go back to look at a program after leaving it for a while.